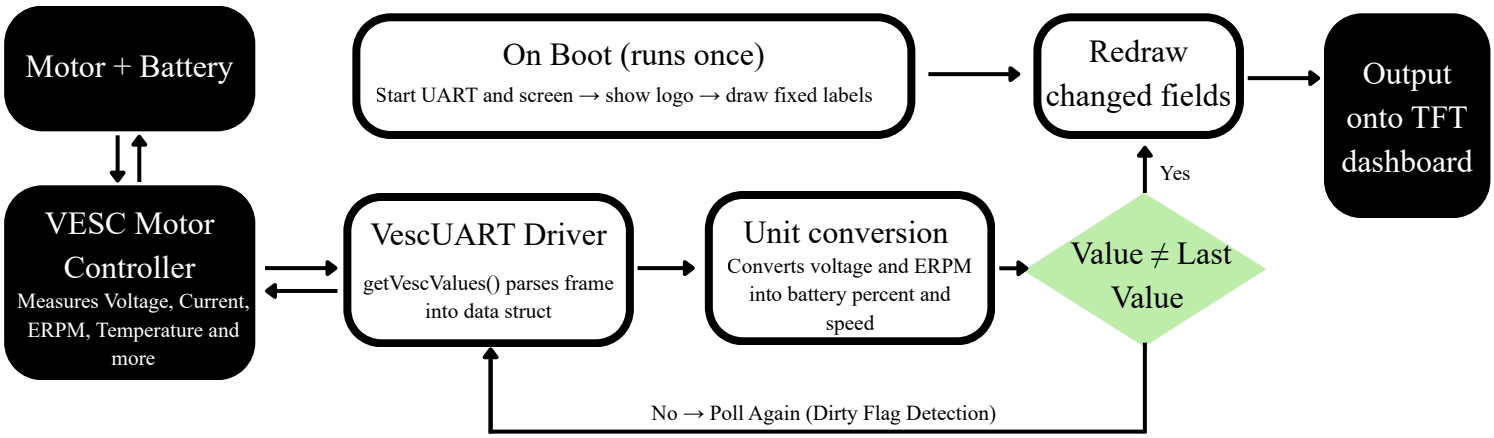
Shivam Khatri - Engineering Portfolio

s25khatr@uwaterloo.ca
github.com/ShivamJKhatri

Electrium Mobility - E-Bike Dashboard Firmware

Sep 2025 - Jan 2026

Firmware Engineer | Design team | Reduced SPI bus writes by 20%



UART Integration

- Set up a hardware UART link (115200 baud) between the ESP32 and the VESC motor controller.
- Each loop, the ESP32 requests a telemetry packet from the VESC and parses 10+ values, including input voltage, motor current, ERPM, duty cycle and MOSFET/motor temperatures.
- Converted raw VESC data into rider units: battery % mapped from pack voltage (30–36 V), and speed in km/h calculated from ERPM.



TFT Dashboard

- Built a 320×480 TFT dashboard in C++ with the TFT_eSPI library, showing voltage, battery % and speed.
- Designed a 6-level battery icon that turns yellow below 20% and red below 5%.
- Stored the logo and battery icon as RGB565 bitmaps in flash (PROGMEM) and wrote a pixel blitter, so no SD card or file system is needed.
- **Dirty-flag rendering**: every field keeps its last drawn value and only redraws when it changes; the battery icon tracks its fill level separately. This cut **SPI bus writes by about 20%**

Debug and Bring Up

- Documented a bring-up checklist (baud, swapped TX/RX, common ground, VESC UART enabled, TFT pins, RGB/BGR order) and wrote bench-test code that steps simulated values to exercise the UI without the motor running.

StabiliKnee - Predictive Orthosis System

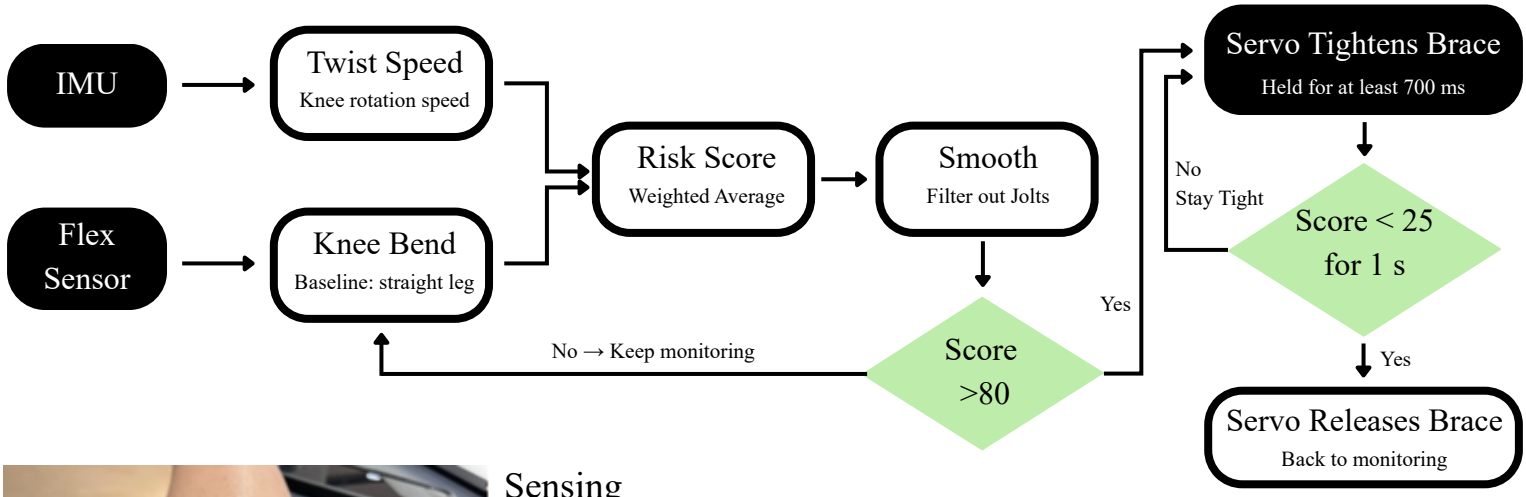
Feb 2026

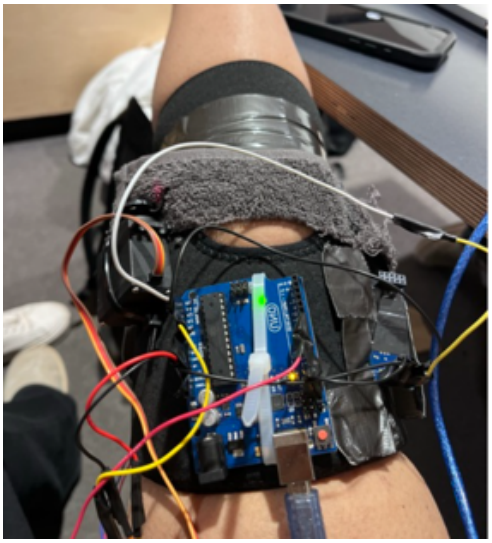
Embedded and Electrical lead | ~70% reduction in simulated dislocations | 3rd at Canada's Largest Hardware Hackathon

Overview: A body-worn device that watches knee motion with an IMU, flags a dangerous movement within 300 ms, and drives a servo to push against it before the kneecap dislocates. Existing braces only give passive support.

Problem

- A teammate has recurring patellar dislocations and was once bedridden for a month after one.
- Off-the-shelf braces are passive: they add support but cannot react to the specific motion that causes a dislocation.



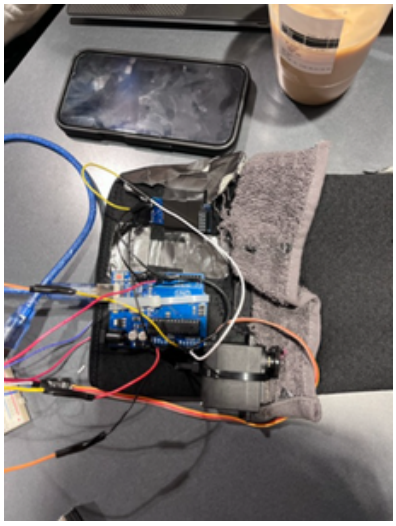


Sensing

- Read a 3-axis IMU over I2C with a 6-byte burst read to get X/Y/Z angular velocity in one transaction.
  - $\omega = \sqrt{\omega_x^2 + \omega_y^2 + \omega_z^2}$
- Configured the gyro's registers directly and converted raw counts to °/s with the datasheet scale of 14.375 LSB per °/s.
- Read a flex sensor across the knee through a voltage divider (10 kΩ resistor) to measure knee bend.
  - $s = \frac{|\text{flex} - \text{flex}_{\text{baseline}}|}{1023}$
- Ran a startup calibration: averaged 1,500 gyro samples to remove each axis's zero-rate bias, and recorded a straight-leg flex baseline.

Why the thresholds work this way

- Two levels (tighten above 80, release below 25) stop the servo from rapidly switching on and off when the score hovers near one value.
- The 700 ms hold and 1 s release delay keep the brace from letting go in the middle of a dangerous movement.



My Contributions

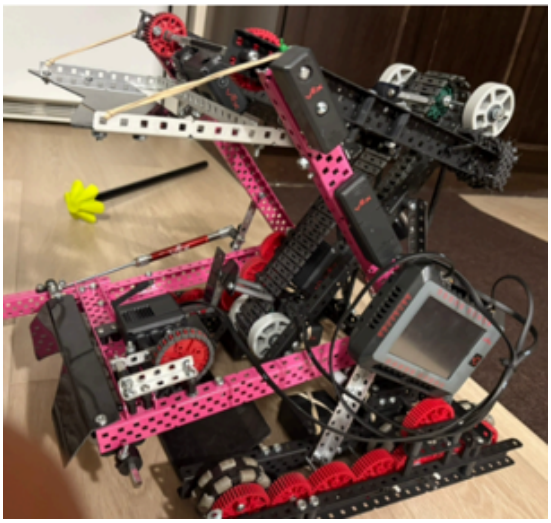
- Reverse-engineered a 15-year-old gyro with no breakout board or docs: mapped its pinout from the chip datasheet, wired it to the Arduino over I2C, and fixed its register addressing through testing.
- Owned all electronics and sensor code: wiring, servo power calculations and integration, plus the filtering and threshold logic that triggers the brace.

VEX Robotics

2022 - 2025

Engineering and Documentation Lead | 4th in province of Alberta | 7+ Awards

Overview: A competition robot that intakes rings, scores them on mobile goals and wall poles, and clamps mobile goals pneumatically, with both driver-controlled and fully autonomous routines.

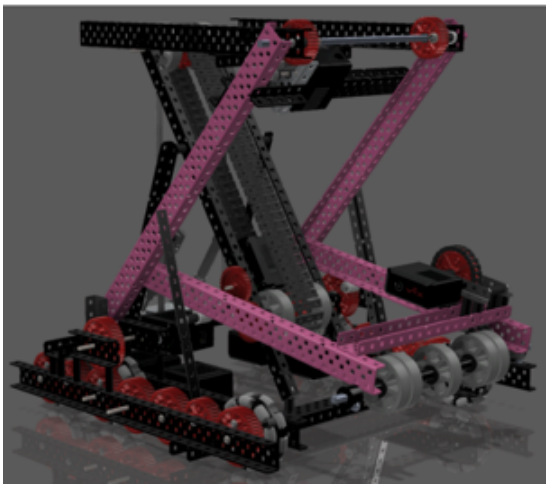


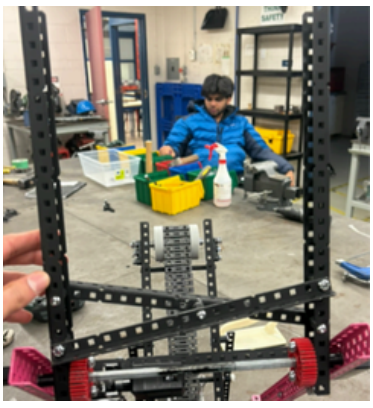
Software

- Autonomous driving: robot drives itself for the first 15 s
  - How it works: target → PID → motors → odometry → correct
  - Hand-tuned PID: lands within 0.3 in of a 25 in target and 0.4° of a 90° turn
  - Wrote 4+ routes (red/blue sides) plus a skills route
- One-button scoring arm: driver scores with a single press
  - How it works: button → preset angle → PID → arm motor
  - A rotation sensor tracks the arm angle so it stops on target
  - 3 presets: rest (2°) → hold ring (40°) → score (160°)
- Colour sorter: keeps only our alliance's rings
  - How it works: optical sensor → reads colour → keep/reject

Mechanical

- Drivetrain: wheels and motors that move the robot
  - How it works: 6 omni wheels, geared to 600 RPM, with gears mounted outside the frame for a tighter turning circle
  - Initial/Final times: 11.1 s to drive the test course →7.7 s
- Intake: picks rings up off the floor and loads them onto the goal
  - How it works: rubber flex-wheels float up and down to grab rings at any angle, then a chain of hooks carries each ring up and drops it onto the goal
  - Initial/Final times: 10.0 s to pick up 5 rings → 5.3 s
- Scoring arm: lifts rings onto the posts along the field wall
  - How it works: a motor geared down 3:1 for extra lifting strength swings the arm, and a mesh holder grips the ring
  - No hard stop, so the arm spins a full 360°: one arm scores on every post and can knock over the opponent's goals





Problems Solved

- Arena lighting broke the colour sorter. Purple dome lights read as both red and blue, so good rings were ejected. Fix: only sort when a ring is actually detected.
- Autonomous missed at one event. Traced it to eyeballing the start position and clamping goals at an angle. Fix: alignment jig and goal aligners.
- Lady brown couldn't grip rings. Flipped the arms outward and added a mesh holder, and it became one of the most reliable parts of the robot.

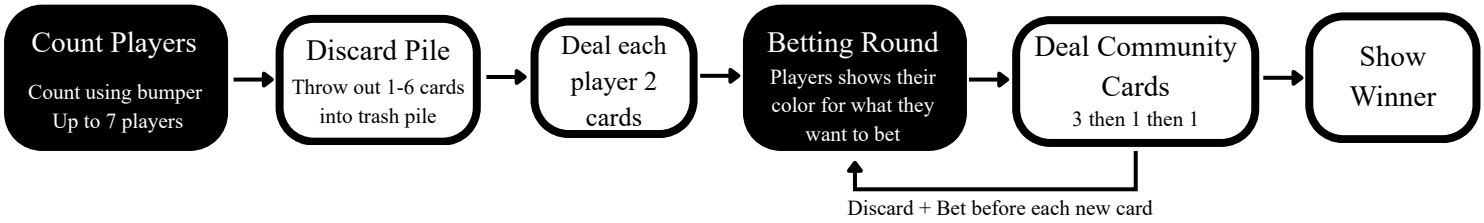
PokerBot - Autonomous Card Dealing Robot

Sep - Dec 2025

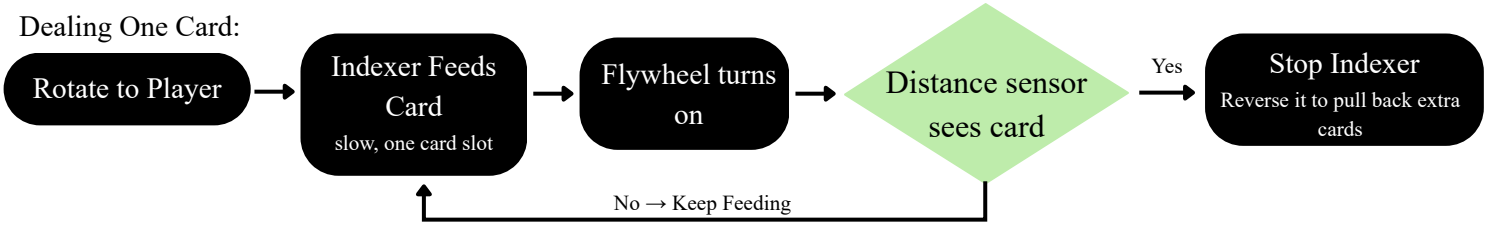
Engineering Lead | 7 players supported | ± 2 cm placement accuracy at 20 cm range

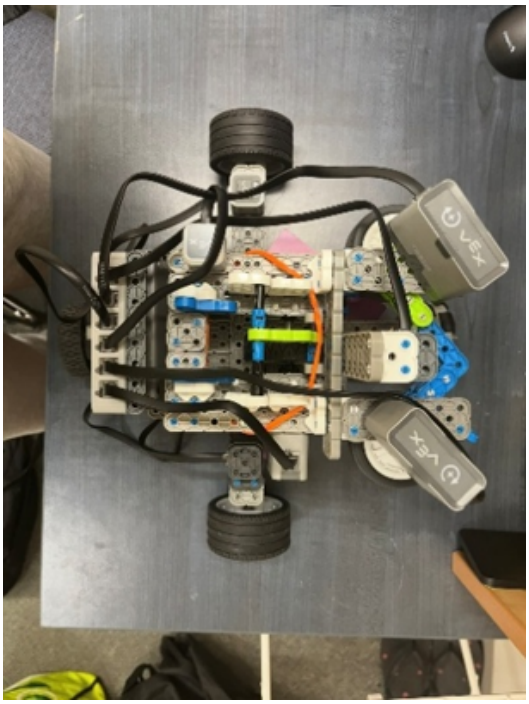
Overview: A robot that deals cards to up to seven players without a human dealer, so nobody can cheat with sleight of hand. It handles player count, betting, a randomized discard pile and community cards across a full round.

Game flow:



Dealing One Card:





Mechanical

- Rotating base: turns the whole robot to face each player
  - How it works: a vertical motor drives a gear under the top half. Wheels on all 4 sides take the weight off the centre gear.
  - The motor wire would wrap around the shaft, so rotation is limited to 180° each way. The deal spreads players across 270°
- Card holder: feeds exactly one card at a time
  - How it works: the deck sits angled up toward a one-card-wide slot. A rubber-band tensioner presses the tip of the deck down so the bottom card always grips the indexer wheel.
- Card launcher: shoots cards face-down across the table
  - How it works: indexer wheel → slides one card forward → two flywheels → card launched
  - The flywheels sit far ahead of the indexer, so even if two cards slip out, only one gets gripped and shot.

Problems Solved

- Robot slid around while dealing: added rubber feet to the base for grip.
- First launcher shot a second card before the first was out: split the job into an indexer that feeds and flywheels that shoot.
- Colour sensor gave random readings: limited bets to three clear colours and a fixed $10 raise.

